

Dynamic Pricing: Main Project Report

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1 Introduction

This project focuses on developing a dynamic pricing strategy in a competitive environment with unknown demand. Each team acts as a seller that must iteratively set prices over multiple periods while learning from observed demand and the pricing of competitors. The objective is to design an adaptive algorithm that maximizes expected revenue by leveraging historical data and responding effectively to market conditions.

2 Literature Review

Dynamic pricing in competitive environments is commonly studied as a sequential decision-making problem under uncertainty, often modeled using contextual bandits. In this framework, actions are selected based on observed context while learning demand from data. The work of Philippe Rigollet and Assaf Zeevi [1] provides a foundation for such approaches, where contextual information, such as competitor behavior, can be incorporated into pricing decisions, and the action space is frequently discretized.

A central challenge in these problems is balancing exploration and exploitation. Thompson Sampling is a widely used method for addressing this tradeoff, as it incorporates uncertainty by sampling from a posterior distribution over model parameters. As discussed by Tor Lattimore and Csaba Szepesvári [2], this approach enables adaptive decision-making that naturally transitions from exploration to exploitation as more data becomes available. Extensions to contextual settings with linear models have been studied by Shipra Agrawal and Navin Goyal [3], where demand or reward is modeled as a function of observed features.

In dynamic pricing applications, demand may evolve over time due to changing market conditions and competitor behavior. Prior work emphasizes the importance of adapting to such changes by prioritizing recent observations. For example, Adel Javanmard [4] highlights the value of using recent data in non-stationary environments. Additionally, modeling demand as a function of contextual variables is well-supported in the literature, motivating the use of features that capture both own pricing decisions and the competitive landscape.

3 Approach Overview

Our strategy integrates four main components, drawing on ideas from the literature discussed above:

1. **Forced exploration** to gather informative data,
2. **Feature engineering** that incorporates competitor price summaries and lagged demand information,
3. **Bayesian linear regression to estimate demand** under current market conditions,
4. **Thompson Sampling** over a discrete price grid to **select the revenue-maximizing price**.

Together, these components allow us to learn the demand over time while continuously adapting our pricing decisions to recent observations and competitive behavior. In the following sections, we provide a detailed description of each component and the reasoning behind the modeling choices.

4 Data

Data is observed sequentially over multiple time periods. At the end of every period, we observe our own realized demand corresponding to the price we set, as well as the prices chosen by all competing teams. Historical prices for all teams are publicly available, while each team’s demand history remains private.

5 Forced Exploration

In the early stages of the simulation, limited data makes it difficult to estimate demand accurately, so exploration plays a crucial role. We begin by selecting prices uniformly at random over the interval $[1, 100]$ for the first 30 periods, which ensures that we observe demand responses across a wide range of prices. After this initial phase, we transition to a more structured exploration. For the next 50 periods we cycle through a predefined grid of candidate prices, effectively testing every other integer value in $[1, 100]$. This guarantees coverage of the entire price space in a systematic way. Even after the warmup phase, we also maintain a small probability of random price selection in order to prevent the strategy from converging prematurely to suboptimal pricing decisions.

6 Feature Engineering

After the initial 80-round exploration phase, we transition to using the Bayesian linear regression and Thompson Sampling strategy. To effectively model demand, we construct a feature vector that captures both our pricing decisions and the competitive environment; it includes the following components:

- Baseline demand (intercept term): we include a constant feature equal to 1 so that the model can learn a baseline level of demand independent of prices and other factors.
- Our own price p_t .
- Nonlinear term p_t^2 that captures the fact that demand is more sensitive at lower prices.
- Competitor price summaries:
 - Average competitor price,
 - Minimum competitor price,
 - Maximum competitor price,
 - Standard deviation of competitor prices.
- Relative positioning features:
 - Ratio of own price to average competitor price,
 - Ratio of own price to minimum competitor price,
 - Fraction of competitors priced below our price.
- Temporal features:
 - Previous demand,
 - Previous own price.

These values are all normalized (divided by 100) to keep them in reasonable range for regression. Together, these features allow the model to capture both static pricing effects and dynamic market behavior.

7 Estimating Demand using Bayesian Linear Regression

We model demand using a Bayesian linear regression framework. Specifically, we assume that demand can be expressed as a linear function of the constructed features plus a Gaussian noise term. Formally, we write

$$d_t = x_t^\top \theta + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

where x_t is the feature vector as described above and θ is an unknown parameter vector. To incorporate uncertainty and enable learning, we place a Gaussian prior on the parameter vector θ . Formally,

$$\theta \sim \mathcal{N}(0, \sigma_0^2 I)$$

. Rather than using all historical data for training the demand model, we estimate the model using a rolling window of the most recent 80 observations. This focuses the model on the most relevant data and avoids relying on outdated information. Within this window, we also apply a weighting scheme that assigns greater importance to more recent samples. Specifically, we use linearly increasing weights, ranging from 0.7 for older observations to 1.3 for more recent ones. As a result, recent observations have a larger influence on the regression, while older observations are slightly downweighted.

Using this weighted data, we update the prior to obtain a posterior distribution over θ , which provides both point estimates of the parameters and a measure of uncertainty captured by the covariance matrix. For numerical stability, we include a small ℓ_2 (ridge) regularization term when estimating the model parameters. This posterior distribution is then used in the Thompson Sampling step to guide pricing decisions. This probabilistic framework naturally balances exploration and exploitation, as greater uncertainty leads to more exploration, while increased confidence results in more consistent pricing decisions.

8 Pricing Strategy using Thompson Sampling

Given the estimated demand model, we use Thompson Sampling to choose a price that maximizes revenue. This works as follows:

1. Sample parameter vector θ from the posterior distribution.

Note, because we assume a Gaussian prior on θ and Gaussian noise in the demand model, the posterior distribution over θ is also Gaussian. This allows us to sample parameter vectors directly from a multivariate normal distribution in the Thompson Sampling step.

2. For each candidate price p in the predefined price grid, compute predicted demand $\hat{d}(p) = \mathbf{x}(p)^\top \theta$, where $\mathbf{x}(p)$ is the feature vector constructed for price p and θ is the sampled demand coefficient vector.

The price grid contains 199 prices in $[1, 100]$ spaced by 0.50. Since we do not observe competitors' future prices, which are needed to construct the feature vector for demand prediction, we approximate the competitive environment using recent observations. We do this by using a blend: 60% weight on the most recent observed prices and 40% weight on the 10-round moving average. This balances responsiveness to recent competitor actions with stability against short-term price fluctuations.

3. Choose the price that maximizes expected revenue:

$$p^* = \arg \max_p p \cdot \hat{d}(p).$$

This approach allows us to naturally incorporate uncertainty in the demand estimates, as different sampled parameters may lead to different pricing decisions. As a result, the strategy balances the need to explore uncertain regions of the price space with the goal of maximizing revenue.

9 Adaptive Model Updating

In each period, we re-estimate the model using a rolling window of recent observations rather than updating the posterior using all past data. This is because the demand environment is dynamic, and older observations may no longer reflect current market conditions due to changes in competitor behavior or demand patterns. By focusing on recent data and downweighting older observations, the model can adapt more quickly to changes, avoiding over-reliance on outdated information.

10 Alternative Approaches

An alternative to Thompson Sampling is to select prices based solely on the posterior mean of the demand model, effectively minimizing prediction error on historical data. While this approach is simpler, it does not account for uncertainty in the parameter estimates and therefore tends to converge prematurely to suboptimal pricing decisions.

In contrast, our use of Thompson Sampling incorporates uncertainty by sampling from the posterior distribution of demand coefficients. This leads to more effective exploration of the price space, particularly in early stages or when data is limited. As more observations are collected and uncertainty decreases, the method naturally shifts toward exploitation, resulting in more stable pricing decisions.

Given the dynamic and competitive nature of the environment, explicitly accounting for uncertainty is critical. For this reason, Thompson Sampling provides a more robust and adaptive approach compared to purely deterministic strategies based on point estimates.

11 Conclusion

Overall, our approach combines data-driven demand estimation with a competitive pricing framework. By incorporating both own-price effects and competitor behavior, and by leveraging Bayesian learning with Thompson Sampling, the strategy is able to dynamically adjust prices in response to the market environment. This results in a balanced approach that effectively learns from data while pursuing revenue maximization, making it well-suited for dynamic pricing in competitive settings.

References

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